■ Tiny ImageNet Portfolio: Setup & Infrastructure Guide

Project Stage: Phase 1 - Environment & Baseline
Hardware: Ubuntu 20.04 (Laptop) -> Windows 11 (Remote GPU Desktop)
Stack: Keras 3 (Multi-backend), TensorFlow, PyTorch, Miniconda

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■ Step 1: VS Code (The Command Center)

Install the official Microsoft repository to enable Remote Tunnels.

Update and install dependencies

```
sudo apt update && sudo apt install software-properties-common
apt-transport-https wget -y
```

Import GPG key and add repository

wget -q https://packages.microsoft.com/keys/microsoft.asc -O- | sudo apt-key add -

```
sudo add-apt-repository "deb [arch=amd64]
https://packages.microsoft.com/repos/vscode stable main"
```

Install VS Code

```
sudo apt update && sudo apt install code -y
```

■ Step 2: Miniconda Installation

Miniconda is preferred over venv for AI projects because it manages non-Python dependencies (like CUDA/cuDNN) much more reliably across different OS (Linux vs Windows).

Download Linux installer

wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh

Run installer (Follow prompts, type 'yes' to initialize)

bash Miniconda3-latest-Linux-x86_64.sh

RESTART YOUR TERMINAL NOW

■ Step 3: Environment Setup

We will create a 'mirrored' environment.

Create the environment

```
conda create -n vision_portfolio python=3.10 -y
conda activate vision_portfolio
```

Install Keras 3 and both backends

```
pip install --upgrade pip
pip install tensorflow keras torch torchvision matplotlib numpy pandas tqdm
```

■ Step 4: Project Structure & Data Fix

1. **Create Folders:**
```bash
mkdir -p ~/TinyImageNet_Project/{data,models,notebooks,src}
cd ~/TinyImageNet_Project
touch src/data_loader.py src/models.py src/train.py README.md .gitignore
```

2. **Download Tiny ImageNet:**
```bash
cd data
wget http://cs231n.stanford.edu/tiny-imagenet-200.zip
unzip tiny-imagenet-200.zip
rm tiny-imagenet-200.zip
```

3. **The Validation Script (Run this in src/data_loader.py):**
Paste the reorganization logic provided in the chat to move validation images into class-named subfolders.

■ Step 5: Remote Bridge (For Day 8)

When you reach your Windows Desktop:
1. Open PowerShell: `code tunnel`
2. On Laptop VS Code: Install "Remote - Tunnels" extension.
3. Click Blue Icon (Bottom Left) -> Connect to Tunnel.